GWTC forecasting

An illustrated primer for your project

The central question in the latest student plan is:

> ***Does an additional high-mass excess, inferred from earlier gravitational-wave observations, improve predictions for later detections beyond a comparable model that already allows a broadly bending mass distribution?***

This is more careful than asking whether "a peak fits the data better than a power law." A sufficiently rigid power law can lose simply because it cannot bend. Your revised comparison asks whether an **additional concentration of black holes earns its place by predicting new observations**.

The earlier project plan already emphasized learning from older catalogs and testing disputed mass-distribution features on later observations. The refurbished version sharpens that into a matched model comparison, with realistic measurement uncertainty, run-dependent selection, and tests of whether the experiment can actually distinguish the alternatives.[1]

There are three ideas to keep in mind throughout:

Astrophysics produces a population. The detector produces a biased, uncertain sample. Your analysis asks which population model best predicted that sample.

The formation illustration linked below is from Mandel and Farmer's *Stellar palaeontology*. The three plots below are numerical teaching examples calculated for this explanation, **not results from your students' analysis**.[2]

Formation illustration (online)

Open the original formation illustration

Three possible mechanisms for the merger of black holes, from *Stellar palaeontology*.[2] The original third-party image is linked rather than embedded because its image file could not be retrieved during PDF conversion. The three locally available teaching plots are embedded in the pages that follow.

Conversion note. The text, scientific claims, and cited source links are retained from the preceding write-up. This PDF is a format conversion, not a new literature review.

1. What population are we trying to understand?

Your immediate target is the population of **merging binary black holes**, not all black holes in the Universe.

For each binary, we conventionally label the more massive component m_1, the less massive component m_2, and their ratio

$$q = \frac{m_2}{m_1} \leq 1.$$

The **primary-mass distribution** describes how often merging systems have different values of m_1. Here, m_1 is the mass of one black hole **before merger**, not the mass of the final remnant. These are the conventions used in the LVK population analyses.[3]

Imagine observing thousands of mergers with perfect instruments. You could construct a distribution answering:

> *Among all the binaries that merge, how common are primaries of* $10\,M_\odot$, $25\,M_\odot$, *or* $50\,M_\odot$*?*

That distribution contains information about how stars make black holes, how binaries interact, and whether some black holes have merged before.

But your students do not observe this ideal distribution directly. Their task is to infer it from incomplete and noisy evidence, then test whether the inferred structure predicts genuinely new events.

Why a mass feature is physically interesting

A localized excess means that a particular mass range is produced, paired into binaries, or brought to merger more efficiently than neighboring masses.

One possible mechanism is **pulsational pair instability**: stellar evolution can remove mass in a way that creates structure in the remnant-mass distribution. Such considerations motivated population models containing both a declining continuum and an additional mass component. However, a fitted Gaussian component is a phenomenological description, not a unique identification of that mechanism.[4]

That distinction is important:

> ***Finding a reproducible mass feature establishes something about the population. Identifying its formation mechanism requires additional evidence.***

2. Where do merging black holes come from?

Read each column of the opening illustration **from top to bottom**. The original image is available through the link on the opening page.

Left: two stars evolve together

Two massive stars begin as a binary. As one expands, mass transfer and sometimes a shared gaseous envelope can change their separation dramatically. In a common-envelope phase, orbital energy can help expel the envelope while the orbit shrinks.

The resulting black holes can therefore be close enough to merge through gravitational-wave emission. The key physical challenge is making a sufficiently compact binary without destroying the system.[5]

Middle: stars remain compact

In the chemically homogeneous channel, rapid rotation mixes stellar interiors. Under suitable conditions, the stars remain relatively compact rather than developing the usual extended structure.

This provides another route from a close stellar binary to a close black-hole binary, without requiring the same envelope-ejection history.[5]

Right: gravitational encounters assemble the binary

In a dense stellar environment, black holes can encounter other stars and black holes. Exchanges and subsequent interactions can assemble and tighten binaries.

The two eventual merger partners need not have been born as a stellar binary together.[5]

A further possibility: a black hole merges more than once

A merger remnant can become a member of another binary. This is a **hierarchical merger**.

Mass and spin together are informative here. The GWTC-5 population analysis reports rapidly spinning subpopulations consistent with hierarchical-merger signatures, while explicitly discussing formation-pathway interpretation rather than treating it as uniquely established.[3]

How this connects to your project: different formation histories may leave different population patterns. Your current mass-focused experiment tests whether one proposed pattern survives new observations. It does **not**, by itself, measure the fraction formed in each environment.

3. What exactly are the students comparing?

The current literature makes your more careful framing worthwhile. The GWTC-5 population analysis reports a peak around $10\,M_\odot$, but describes the feature around $35\,M_\odot$ as more consistent with a **change of slope** than a distinct local peak.[3]

A bend and an additional excess are different claims.

A **bend** says:

> *The abundance falls with mass, but its rate of decline changes.*

An **additional excess** says:

> *After allowing that broad change, a particular mass region still needs extra probability.*

An illustrated matched comparison

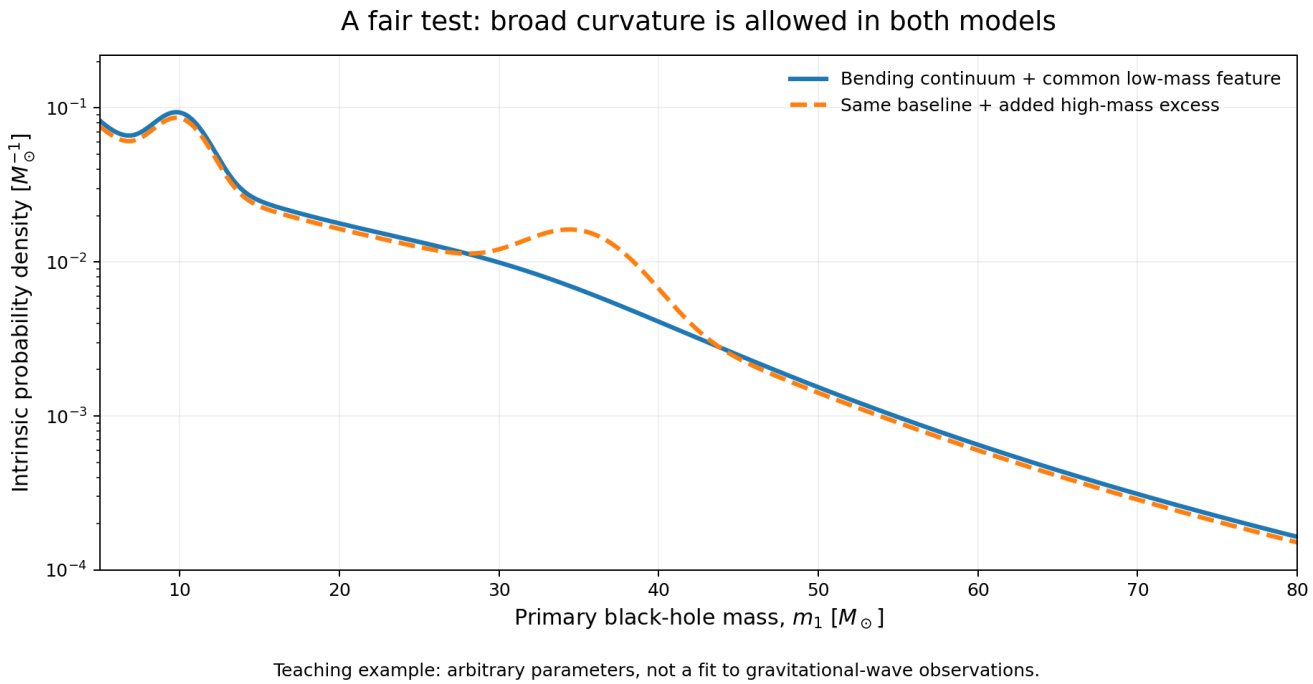


Figure 1: Two illustrative population models, both allowing a broadly bending continuum, with only one containing an additional high-mass excess. Teaching example, not a fit to gravitational-wave observations.

In this teaching example, both models share a low-mass feature and a broadly bending continuum. Only the dashed model adds a component near $35\,M_\odot$.

A schematic representation is

$$p_{\text{excess}}(m_1) = (1-f)\,p_{\text{baseline}}(m_1) + f\,g(m_1;\mu,\sigma),$$

where g is a normalized localized component and f is its population fraction. This equation illustrates the comparison; it is **not a claim about the exact functions or priors in the refurbished repository**.

Notice something subtle in the plot: adding probability near $35\,M_\odot$ reduces probability elsewhere because the distribution must still integrate to one.

That creates a real predictive obligation. The additional component cannot simply improve the fit locally without consequences:

> ***To predict more binaries near the proposed feature, the model must predict relatively fewer somewhere else.***

A later catalog can reward or penalize that redistribution.

Also, an added component need not produce a conspicuous local maximum in the total distribution. A weak or broad excess may appear only as a shoulder. The scientific target is its **incremental predictive value**, not whether the plotted curve looks sufficiently "peaky."

4. What does a gravitational-wave observation actually tell us?

The detector measures a time-dependent strain signal, not a black-hole mass directly. Waveform analysis turns the signal into a probability distribution over masses, spins, distance, orientation, and other parameters.[6]

Consequently, one event should be thought of as a **cloud of plausible binary configurations**, not a point on the mass axis.

For example, an event might allow substantial probability across both the proposed excess and the neighboring continuum. Its contribution to model comparison must average over that uncertainty.

Replacing every event with its posterior median would throw away precisely the information needed to decide whether the event really distinguishes the models.

Source-frame versus detector-frame mass

Cosmological redshift stretches the waveform:

$$m_{\text{detector}} = (1+z)m_{\text{source}}.$$

Thus, a hypothetical $30\,M_\odot$ source at $z = 0.5$ has the same redshifted mass scale as $45\,M_\odot$ in the detector frame. Inferring the source-frame mass requires treating distance, redshift, and the assumed cosmology consistently.[7]

This is why "we loaded the mass column" is not enough. The students must know which mass convention they loaded and which prior generated the posterior samples.

What the students can reuse

The analysis can reuse public event-level posterior samples rather than repeating strain-level parameter estimation. The statistical operation is called **posterior recycling** or **reweighting**: use the existing samples to evaluate how well a proposed population explains each event.[8]

This is what makes a focused population-prediction project practical without first building an entire gravitational-wave analysis pipeline.

5. The detector changes the apparent population

This is the closest connection to your high-energy-astrophysics experience.

You would not compare an intrinsic neutrino spectrum directly with reconstructed energies while ignoring effective area and energy resolution. The analogous mistake here is comparing an intrinsic black-hole mass distribution directly with a histogram of catalog mass estimates.

The conceptual sequence is

$$\text{all mergers} \longrightarrow \text{detected mergers} \longrightarrow \text{uncertain measurements}.$$

A smooth population can produce a concentration of detections

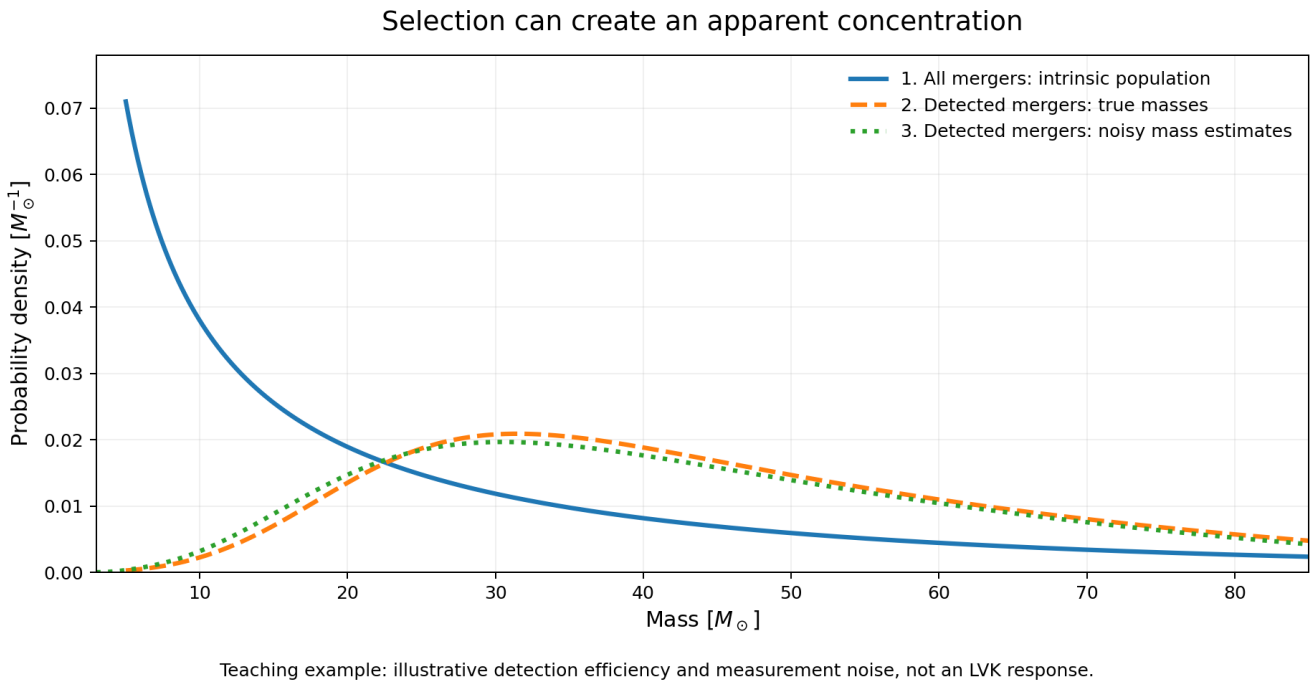


Figure 2: An illustrative smooth intrinsic population develops an apparent concentration after detection selection; measurement uncertainty further changes the observed distribution. The detection efficiency and measurement noise are illustrative, not an LVK response.

Here the intrinsic population decreases smoothly. Yet the detected population has a broad maximum because low-mass systems are difficult to detect in this constructed example.

The dotted curve adds measurement noise.

Nothing in the intrinsic population contains the apparent concentration seen among detections.

This is an illustration, not an explanation of the actual $35\,M_\odot$ feature. Real sensitivity depends on multiple parameters, and increasing mass does not improve detectability indefinitely because the signal can move out of the instrument's sensitive frequency range.[8]

The selection function

Let θ denote an individual binary's parameters and Λ the parameters describing the population.

For observing run r, define

$$\alpha_r(\Lambda) = \int p_{\mathrm{det},r}(\theta)\, p(\theta \mid \Lambda)\, d\theta.$$

This is the fraction of the specified population expected to pass the analysis's detection selection.

The same intrinsic population can give different detected distributions in different observing runs. Therefore, **freeze the astrophysical model for the prediction test, but use the appropriate detector selection for each run.** Accounting for uncertain measurements and selection together is central to hierarchical population inference.[9]

A useful intuition is that O3 and O4 are not two universes whose stellar evolution changed over a few years. They are new samples observed through different instrumental conditions, potentially reaching different distances.

6. The two calculations at the heart of the students' work

The latest division of responsibilities makes sense because the analysis depends on two complementary integrals.

Student A's central calculation: how compatible is this event with this population?

For event i,

$$\ell_i(\Lambda) = \int p(d_i \mid \theta)\, p(\theta \mid \Lambda)\, d\theta.$$

Here d_i is the observed event data.

Intuitively, consider every binary configuration that could have produced the event. Give it more weight when the proposed population predicts it more frequently.

Using equally weighted posterior samples $\theta_i^{(k)}$,

$$\ell_i(\Lambda) \propto \frac{1}{K_i} \sum_{k=1}^{K_i} \frac{p(\theta_i^{(k)} \mid \Lambda)}{\pi_{\mathrm{PE},i}(\theta_i^{(k)})}.$$

The denominator removes the prior used in the original parameter-estimation analysis. Otherwise, that prior would be mistaken for population evidence. The omitted event-evidence factor cancels when comparing population models using the same event analysis.[8]

A practical interpretation is:

> ***Remove the assumptions used to produce the event samples, then ask how the proposed population weights those same possibilities.***

For weighted posterior samples, the averaging must retain their weights.

Student B's central calculation: how many members of this population would be detected?

Sensitivity studies inject simulated signals and record which are recovered.

In the simplest importance-sampling setup,

$$\hat{\alpha}_r(\Lambda) = \frac{1}{N_{\mathrm{generated}}} \sum_{j \in \mathrm{recovered}} \frac{p(\theta_j \mid \Lambda)}{g_r(\theta_j)},$$

where g_r is the distribution used to generate the injections. Actual release products can have additional weighting and sampling conventions that must be respected. Injection-based sensitivity estimates are themselves Monte Carlo quantities with finite precision.[10]

A basic test makes the normalization memorable:

Suppose 10,000 signals are generated, 1,000 are recovered, and the injection distribution exactly matches the target population. Every importance weight is one, so

$$\hat{\alpha} = \frac{1000}{10000} = 0.1.$$

Dividing by the number **recovered** would incorrectly give one.

That deliberate error is an excellent teaching test because the code may run perfectly while the statistical interpretation is completely wrong.

Why the early quadrature checks matter

In a simple, low-dimensional example, the students can evaluate these integrals directly with numerical quadrature and compare the result with sample-based estimates.

This tests the scientific calculation against an independent numerical route, rather than asking whether two pieces of code share the same assumptions or mistake.

7. What does “fit the old catalog, then freeze” mean?

First, combine event compatibility with selection.

For a shape analysis conditioned on the number N_r of detections in each run, the likelihood has the form

$$\mathcal{L}_r(\Lambda \mid N_r) \propto \frac{\prod_{i\in r} \ell_i(\Lambda)}{\alpha_r(\Lambda)^{N_r}}.$$

The selection denominator accounts for the fact that the catalog contains detected systems rather than arbitrary draws from the intrinsic population. This is distinct from a full rate analysis, which also models the number of detections.[9]

The earlier catalog then gives a posterior distribution

$$p(\Lambda \mid D_{\text{train}}, M)$$

for each model M.

Freeze the uncertainty, not merely the best fit

Suppose the earlier data allow a feature centered anywhere from roughly 30 to 40 $M_\odot$.

A prediction that fixes the center to 35 $M_\odot$ has discarded uncertainty. It may appear to fail because it was made artificially sharp.

The proper predictive distribution averages over the earlier posterior:

$$\boxed{p(D_{\text{test}} \mid D_{\text{train}}, M) = \int p(D_{\text{test}} \mid \Lambda, M)\, p(\Lambda \mid D_{\text{train}}, M)\, d\Lambda.}$$

Think of the training posterior as an ensemble of plausible populations. Each population predicts a later catalog; the final prediction averages those possibilities.

The frozen objects are the model definition, prior, training information, and specified analysis rules. “Frozen” should not mean pretending that uncertain population parameters are known exactly.

8. How do we decide which model predicted better?

Define the predictive log-score difference

$$\begin{aligned}\Delta S = {} & \log p(D_{\text{test}} \mid D_{\text{train}}, M_{\text{excess}}) \\ & - \log p(D_{\text{test}} \mid D_{\text{train}}, M_{\text{baseline}}).\end{aligned}$$

Positive values favor the additional excess; negative values favor the baseline.

For example, $\Delta S = 2$ means that the later dataset has about $e^2 \simeq 7.4$ times higher predictive density under the excess model, given the training information and the specified analysis.

It does **not** mean a two-sigma detection, nor does it give the probability that the excess is physically real.

Why joint and chronological scores both appear in the plan

There is a subtle distinction between predicting an entire catalog and predicting each event separately from the same frozen reference.

In general,

$$\underbrace{\int \prod_i f_i(\Lambda)\, p(\Lambda)\, d\Lambda}_{\text{joint prediction}} \neq \underbrace{\prod_i \int f_i(\Lambda)\, p(\Lambda)\, d\Lambda}_{\text{product of separately averaged predictions}} .$$

The reason is physical: **all the events share one underlying population**.

An exact chronological decomposition predicts each event using the information available before it:

$$\log p(D_{\text{test}} \mid D_{\text{train}}, M) = \sum_j \log p(d_j \mid D_{\text{train}}, d_{<j}, M).$$

The posterior updates between predictions according to the already specified Bayesian rule. This does not authorize changing the model or inventing a new feature after seeing an inconvenient event.

Separate eventwise scores using the original training posterior are still useful diagnostics, but their sum is not generally the joint score.

What event-influence diagnostics tell you

They help distinguish two very different outcomes:

> *Many later events consistently prefer the added component.*

versus

> *Almost the entire preference comes from one unusual event.*

The second outcome is not automatically invalid. An exceptional event can be genuinely informative. But it calls for a different physical interpretation and careful checks of that event's likelihood and modeling assumptions, not automatic removal.

9. Why calibration and power come before the main claim

These answer different questions.

Calibration: How often does the procedure claim an additional feature when the tested baseline is the truth?

Power: How often does it recognize an additional feature when a specified alternative is the truth?

A procedure can have excellent calibration and almost no power.

A small calculation that makes the distinction concrete

For illustration, suppose two known models predict that 10% or 20% of detected events fall in a specified category. There is no measurement uncertainty in this toy problem.

I calculated an exact binomial test whose false-alarm probability never exceeds 5%.

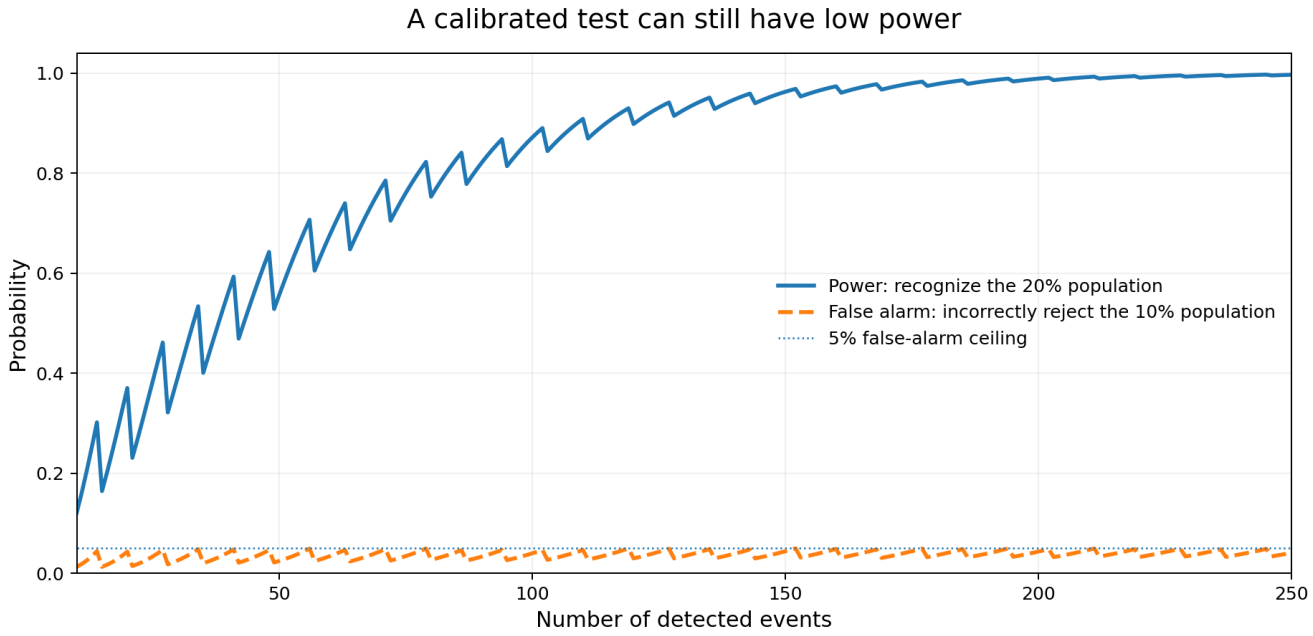


Figure 3: Exact binomial teaching example showing that false-alarm control can remain good while power depends strongly on the number of observations. The category fractions are assumed known; mass uncertainty and training uncertainty are not included.

With 20 observations, the test recognizes the 20% population only about **37%** of the time. With 100 observations, that rises to about **87%**.

The small sawtooth pattern comes from the integer-valued rejection threshold.

These numbers are **not a gravitational-wave sensitivity forecast**. They illustrate why "we did not favor the extra feature" is incomplete without asking whether the experiment could have recognized it.

What the realistic simulations must include

For your project, the mock experiment must reproduce the consequential steps: generate an earlier population sample, create uncertain observations with appropriate selection, infer the training posterior, and score a later mock catalog.

A simulation that gives the analysis exact black-hole masses is testing an easier experiment than the one your students will perform.

Recent work on gravitational-wave population checks shows why apparently satisfactory predictive plots can be misleading when event measurements are weak or prior-dominated. Some failures remain difficult to diagnose because the data themselves carry too little information.[11]

Numerical precision is a separate issue

There are at least three distinct uncertainties:

Population uncertainty concerns what the earlier catalog allows.
Catalog variability concerns which new events happen to occur.
Numerical uncertainty concerns how accurately the integrals and predictive scores are computed.

More posterior or injection samples can reduce the third, not the first two.

For example, a model-dependent fractional selection error ϵ changes the conditional log likelihood approximately by

$$\delta \log \mathcal{L} \simeq -N\epsilon.$$

At $N = 100$, a 1% error can therefore matter at order unity in log likelihood. This simple expansion explains why Monte Carlo precision must be assessed relative to the scientific score difference, not merely by checking that calculations return finite numbers.[10]

10. How to supervise the latest student objectives

The revised workstreams are best understood as a division of scientific responsibility, not merely a division of software modules.

Workstream	What it must establish	A useful question at the meeting
Student A: event likelihoods and population inference	Posterior recycling represents uncertain events correctly, and training inference behaves as expected.	"Can you reproduce a directly integrated event likelihood, and explain the prior correction?"
Student B: selection and predictive qualification	The sensitivity calculation is normalized correctly, and model discrimination is calibrated and informative.	"Can you show both the recovered/generated normalization test and the power of the complete experiment?"
Joint integration	The saved outputs from one part are correctly consumed by the next.	"Can a small catalog pass through the actual analysis and produce an interpretable predictive score?"

The planned **five-event end-to-end exercise** is valuable because it connects the real components. But it is an integration demonstration, not evidence that five events can resolve the astrophysical question.

At each stage, ask for a figure and its scientific interpretation. A report that "the sampler ran" is much less informative than:

> *"This figure shows the range of populations allowed by the earlier data, the later observations they predict, and the uncertainty in the comparison."*

Keep the held-out claim precise

An observing run and a catalog release are not interchangeable. Releases can revise earlier events as well as add new ones; GWTC-4.1, for example, updates O4a results. Event identities and observing periods therefore matter when constructing nonoverlapping training and test sets.[12]

Also, a retrospective O3-to-O4 evaluation is not the same as a forecast registered before O4 results became known. If later literature helped choose the model comparison, describe that exposure honestly. It does not make the analysis useless; it changes the strength and wording of the prediction claim.

What would count as a meaningful outcome?

A positive result would support:

> ***The additional high-mass component learned from earlier observations improves prediction of later detections beyond an already bending baseline.***

A negative result, with adequate demonstrated power, could support:

> ***The tested additional component does not deliver the predictive gain expected if an excess of the specified strength were present.***

An inconclusive result would be:

> ***The available observations do not distinguish these alternatives reliably.***

Neither a predictive win nor a loss immediately identifies or excludes a formation channel. The result concerns the tested population descriptions and their observable consequences.

The central picture to carry into your first meeting

Your students are not simply trying to draw a better black-hole mass histogram.

They are constructing a controlled argument:

$$\text{Earlier observations} \rightarrow \text{uncertain population predictions}$$
$$\rightarrow \text{later detector-level evidence} \rightarrow \text{a calibrated comparison.}$$

The most useful question for you to keep asking is:

> ***"What should the next detected binaries look like under each explanation, and have we demonstrated that our analysis can tell those predictions apart?"***

That question connects the astrophysics, the Bayesian machinery, the detector response, and the students' immediate milestones.

Source links retained from the original write-up

These entries preserve the sources cited in the chat text. Descriptive labels identify their role in the primer; the conversion did not independently verify or update the cited scientific claims. Reference 1 is an internal project-context citation rather than a public document.

[1] **Project context**. Earlier student plan and the project conversation "Advise on Grad Students," as cited in the preceding write-up. The original internal file citation is not available as a resolvable document link in this PDF.

[2] I. Mandel and A. Farmer, *Stellar palaeontology*, *Nature* **547**, 284–285 (2017). Formation-channel illustration. https://www.nature.com/articles/547284a

[3] **GWTC-5 population analysis**, cited for mass-distribution features, mass conventions, and hierarchical-merger interpretation. https://arxiv.org/html/2605.27226v1

[4] **Mass-population modeling and pulsational pair instability**. https://arxiv.org/abs/1801.02699

[5] **University of Birmingham gravitational-wave group: science overview**, cited for binary-black-hole formation channels. https://www.sr.bham.ac.uk/gwgroup/science.php

[6] **Gravitational-wave source inference**, *Physical Review Letters* **116**, 241102. https://link.aps.org/doi/10.1103/PhysRevLett.116.241102

[7] **GWOSC: GW150914 event page**, cited for source-frame and detector-frame mass conventions. https://gwosc.org/events/GW150914/

[8] **Gravitational-wave inference tutorial**, cited for posterior recycling and selection effects. https://arxiv.org/html/1809.02293v9

[9] **Population inference with uncertain observations and selection**. https://arxiv.org/abs/1809.02063

[10] **Monte Carlo precision in population inference**, cited for injection-based sensitivity estimates and numerical uncertainty. https://arxiv.org/abs/2204.00461

[11] **Gravitational-wave population predictive checks**, cited for limitations with weak or prior-dominated event measurements. https://arxiv.org/html/2604.06090v2

[12] **GWOSC: GWTC-4.1 catalog release**. https://gwosc.org/GWTC-4.1/